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Roll No

CE-7005(2)-CBGS

B.E. VII Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Earthquake Resistant Design

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

iii) Draw neat diagram.

1. a) Explain the seismic zoning of India from the beginning and also explain it's significance. 7
b) What do you understand by earthquake? Explain in sufficient details. Also explain the various types of waves generated due to earthquakes. 7
2. a) What is response spectra and explain the importance of in seismic design of a structure. 7
b) The following details are available for a multistory moment resisting building. Compute the lateral forces developed at various levels using I.S. code 1893-1984. Number of stories = 4, constant $\beta = 1.2$ Basic Horizontal seismic coefficient $\alpha_0 = 0.055$ importance factor $I = 1.0$. Performance factor $K = 1.0$ load at each i^{th} floor $W_i = 500$ kN. Height of each i^{th} floor $H_i = 3.0$ m. 7

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3. a) Explain concept of shear walls in high rise building. Explain various types of structural features resisting lateral shear and discuss their performance characteristics. 7
b) What do you mean by Infill walls? How are they useful in making the structure earthquake resistant? Explain various modes / mechanism for failure of infilled frame. 7
4. Explain the recommendations of IS code for detailing of RC beams for ductility. 14
5. a) Explain the different method of seismic retrofitting of various components of reinforced concrete structures. 7
b) Define: 7
 - i) Damping Devices
 - ii) Base isolation system
6. a) What do you understand by intensity and magnitude of the earthquake? 7
b) Explain the analysis of infill wall using equivalent diagonal structure for infill with opening. 7
7. a) Derive general equation of motion for undamped free vibration of an single degree of Freedom System. 7
b) Explain seismograms and accelerograms. 7
8. Explain: 14
 - i) Damped Free Vibration
 - ii) Behavior of tall shear
 - iii) P-A effect

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